Practice Test 01 Sol

Q.1

(a) Field rotates counterclockwise around the origin.

(b) Magnitude: $|\vec{F}| = \sqrt{y^2 + (-x)^2} = \sqrt{x^2 + y^2}$.

At (1,0): $|\vec{F}| = 1$.

(c) Check conservative:

$\frac{\partial M}{\partial y} = 1, \ \frac{\partial N}{\partial x} = -1$ → not equal → **not conservative**.

Q.2

Given $x = \cos t, y = \sin t, 0 \leq t \leq \pi$:

$$f(x, y) = x^2 + y^2 = 1$$

and $ds = \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2 dt} = 1\,dt$.

So,

$$\int_C (x^2 + y^2)\,ds = \int_0^\pi 1\,dt = \pi.$$

Q.3

Parametrize: $x = t, \ y = t, \ 0 \leq t \leq 1$.

Then $d\vec{r} = (1, 1)\,dt$.

$\vec{F} = (y, x) = (t, t)$.

$$\vec{F} \cdot d\vec{r} = (t, t) \cdot (1, 1) = 2t.$$

$$\int_0^1 2t\,dt = [t^2]_0^1 = 1.$$

Q.4

$\vec{F}(x, y) = (-y, x)$, C: unit circle $x = \cos t, y = \sin t$.

$d\vec{r} = (-\sin t, \cos t)\,dt$.

$$\vec{F} \cdot d\vec{r} = (-\sin t)(-\sin t) + (\cos t)(\cos t) = 1.$$

$$\int_0^{2\pi} 1\,dt = 2\pi.$$

Q.5

$\vec{F} = (x, y)$, circle $x^2 + y^2 = 1$.
Outward normal: $\hat{n} = (x, y)$.

$$\vec{F} \cdot \hat{n} = x^2 + y^2 = 1.$$

$$\text{Flux} = \int_C 1\, ds = \text{circumference} = 2\pi(1) = 2\pi.$$

Q.6

$\nabla f = (2xy, x^2 + 3y^2)$.
For any path C from $(1, 0)$ to $(2, 2)$:

$$\int_C \nabla f \cdot d\vec{r} = f(2, 2) - f(1, 0).$$

Compute:
$f(2, 2) = 2^2(2) + 2^3 = 8 + 8 = 16.$
$f(1, 0) = 1^2(0) + 0 = 0.$

Q.7

$\vec{F} = (2xy, x^2 + 2y)$.
$\frac{\partial M}{\partial y} = 2x$, $\frac{\partial N}{\partial x} = 2x$. ✅ Equal → conservative.

Find f:

$$\frac{\partial f}{\partial x} = 2xy \implies f = x^2 y + g(y).$$

Differentiate w.r.t. y:

$$\frac{\partial f}{\partial y} = x^2 + g'(y) = x^2 + 2y \implies g'(y) = 2y \Rightarrow g(y) = y^2.$$

✅ $f(x, y) = x^2 y + y^2$.

Q.8

$\vec{F} = (x^2 - y, x + y^2)$

(a) Check conservative:

$$\frac{\partial M}{\partial y} = -1, \quad \frac{\partial N}{\partial x} = 1 \Rightarrow \text{not equal} \rightarrow \text{not conservative.}$$